

HistoryLand: a Serious Game for History Learning in Primary Education

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Abstract: Research has indicated that games can positively impact learning by providing a potentially motivating and engaging learning environment that can complement traditional education. This is especially important for such domains as History, which students often consider to be one of the least appealing learning subjects. In this paper, we present the design of the prototype online game HistoryLand, along with an evaluation study conducted in the context of a Greek primary school course on Modern History. The aim of this study was to explore the impact of the game on students' immersion, collaboration, knowledge improvement and enjoyment. The findings of the study revealed a positive students' opinion about the HistoryLand game, which was perceived as useful for comprehension of the historical events discussed in class.

Keywords: game-based learning, primary education, History

1. Introduction

Over the past decade, game-based learning has received a lot of attention since games may provide a pedagogically beneficial opportunity for learning and teaching certain subjects in an enjoyable manner to complement traditional instruction (Prensky 2005; Van Eck 2006). Serious games can motivate students, promote active learning, reduce stress, improve focus and concentration, provide deeper understanding, and enhance interpersonal skills and critical thinking (Ellis et al. 2006; Backlund and Hendrix 2013).

However, not all games have a positive impact on students' learning. In fact, some games might even have negative effects, causing students to lose interest (Squire and Barab 2004). Serious games should be developed based on a clearly defined set of educational goals and guidelines (Watson et al. 2011). Also, research has shown that games can yield better results as an assisting medium rather than the main and only source of learning (Wainess 2007).

History is regarded as being one of students' least favorite subjects (Loewen 1995). Typically, students face difficulty memorizing historical facts (Rozita and Zaliza 2005), and tend to get most of their historical knowledge from sources outside the educational system (Letourneau 2006). In view of the above, some researchers decided to explore how History learning could benefit from the utilization of games by means of increasing the levels of students' motivation and engagement (Levesque 2008). Although the studies conducted in this area are rather limited, the existing results seem promising, suggesting that games can help students improve their learning performance and gain deeper understanding of historical concepts and facts (Probert 2013).

Under this scope, we have designed and developed a prototype game, named HistoryLand, as a complementary tool for teaching Modern History in primary education settings. Although the implementation of the game constitutes a work-in-progress, in the following sections we briefly present the game design as well as a pilot evaluation study that elicits students' impressions of the game.

2. A quick overview of the HistoryLand game

HistoryLand was developed using web technologies, such as HTML5, CSS3, JavaScript and MySQL. The design of HistoryLand follows the model for serious educational games proposed by Annetta (2010). The game consists of (a) an individual phase asking students to collect historical item cards and (b) a collaborative phase encouraging students to combine the item cards collected in order to compose a joint story about a historical event.

The first phase consists of three levels, each representing one historical period: the Balkan Wars, World War I and World War II respectively. In every level a set of item cards pertaining one of these three historical periods are presented to the players. Each player is asked to identify and collect the only card that is irrelevant to the historical period of the current level but relates to another key historical event: that of the Asia Minor Campaign in all cases. In each level the player has three lives, which equal to three chances to find the irrelevant item card. The player's score is dynamically affected by the number of their mistakes. After completing the three game levels, a player may have collected up to three item cards. On following the individual game phase described above, players are assigned to dyads and attempt to combine the item cards they have both collected in order to create a joint story about the Asia Minor Campaign. During this collaborative learning activity, a conversational agent provides guidance and scaffolds players' discussions.



Figure 1: A screenshot of the HistoryLand interface (translated from Greek)

3. Pilot study method

3.1 Participants and procedure

A one-shot study has been conducted in the context of a primary school class on 'Modern History'. The experimental activity took place in a Greek school computer lab. The participants were 34 5th grade Greek students (15 males and 19 females) aged 11 to 12 years old.

First, students filled in a student-opinion questionnaire. One week later, students were asked to play HistoryLand. Throughout the first three levels of the game, each student collected up to three item cards displaying historical facts or figures related to the Balkan Wars, World War I or World War II. Thereafter, students collaborated in dyads (17) to compose a story about the Asia Minor Catastrophe of 1922 using the item cards collected. Following this 90-minute activity, students were requested to fill in a post-task questionnaire.

3.2 Data collection and analysis

The pre- and post-task questionnaires included a number of identical questions in order to compare students' opinions on their levels of collaboration, knowledge improvement and enjoyment (Sisson 2011) before and after playing the game. The post-task questionnaire included some additional questions, several of them examining students' immersion. Each of the above dimensions (combined variables) of the post-task questionnaire reflected a high internal consistency coefficient (Table 1, C_α).

All questionnaire items asked students to express their agreement or disagreement with statements using a Likert-scale from 1 (disagree) to 5 (agree). Measures of central tendency were calculated for each questionnaire variable. Moreover, a Spearman's rank correlation coefficient was computed to assess the relationships among the post-task questionnaire variables. Finally, as the normality and the homogeneity of variance criteria were not satisfied, we proceeded to apply non-parametric statistics in order to compare the pre- and post-task questionnaire scores.

4. Results

Tables 1 presents the descriptive statistics calculated for the four post-task questionnaire dimensions as well as some of the post-task questionnaire variables.

Table 1: Post-task questionnaire results

Questionnaire Variables	M(SD)
<i>Enjoyment ($C_\alpha=0.893$)</i>	<i>4.41(0.37)</i>
The game grabbed my attention	4.76(0.76)
The game provided content that stimulated my attention	4.67(0.83)
Generally speaking, I was able to remain concentrated in the game	4.38(0.87)
I cannot recall any distraction from the task	3.82(1.54)
<i>Immersion ($C_\alpha=0.813$)</i>	<i>4.18(0.64)</i>
I forgot about time passing while playing the game	4.73(0.78)
I became unaware of my surroundings while playing the game	3.35(1.59)
I temporarily forgot worries of everyday life while playing the game	3.76(1.54)
I felt highly involved in the game	4.85(0.49)
<i>Collaboration ($C_\alpha=0.947$)</i>	<i>4.73(0.67)</i>
Collaboration in the game helped me learn	4.73(0.60)
The game helped me collaborate with my classmate	4.82(0.75)
Chatting with my partner helped me understand the subject better	4.58(0.73)
I enjoyed chatting with my partner	4.79(0.58)
<i>Knowledge improvement ($C_\alpha=0.903$)</i>	<i>4.59(0.26)</i>
The game helped me improve my knowledge	4.41(1.19)
I understood the main concepts presented	4.26(1.09)
I tried to apply any prior knowledge to the game	4.85(0.60)
The game motivated me to consolidate knowledge gained in class	4.82(0.71)

A Wilcoxon Signed-ranks test was conducted only for the set of questionnaire items that was identical in the pre- and post-task questionnaires. Three significant differences were reported. Firstly, there was a significant difference in the scores of a question eliciting students' opinion about the pedagogical value of computer-based activities in class ($Z=-1.97$, $p=0.04$, $r=-0.24$). More specifically, after the experimental activity, students rated computer-supported learning activities as significantly more useful after playing the game ($M=3.41$, $SD=1.42$), as compared to their pre-task questionnaire scores ($M=2.84$, $SD=1.46$).

Secondly, the experimental activity stimulated a statistically significant change in students' perception of their concentration levels ($Z=-2.61$, $p=0.00$, $r=-0.32$). More specifically, students regarded their

levels of distraction during the HistoryLand activity ($M=2.38$, $SD=1.59$) as significantly lower as compared to a typical History lesson ($M=3.25$, $SD=1.52$).

Thirdly, an additional Wilcoxon Signed-ranks test revealed a significant effect of the particular game-based learning activity on altering students' attitudes toward collaboration in class ($Z=-3.57$, $p=0.00$, $r=-0.43$). In particular, students' preference for working in groups over working individually when participating in learning activities was found to be substantially higher after playing the HistoryLand game (pre-task: $M=3.65$, $SD=1.54$; post-task: $M=4.65$, $SD=0.92$).

Table 2 illustrates the statistically significant results of the Spearman's rank correlation test revealing the linear correlations between post-task questionnaire variables.

Table 2: Significant correlations emerging from the post-task questionnaire

Variable 1	Variable 2	r	p
I forgot about time passing while playing the game. (immersion)	The game provided content that stimulated my attention. (enjoyment)	0.914	0.000
I forgot about time passing while playing the game. (immersion)	The game helped me improve my knowledge. (knowledge improvement)	0.793	0.000
Collaboration in the game helped me learn. (collaboration)	The game helped me improve my knowledge. (knowledge improvement)	0.887	0.000

5. Conclusions and future work

In this paper we have presented the HistoryLand game along with a pilot evaluation study. The main goal was to explore the user acceptance of HistoryLand when used as a complementary tool for teaching Modern History in Greek primary education. The preliminary analysis of the results revealed a favorable students' perception of the game. Students stated enjoying playing the game, which aroused their interest. Furthermore, most of the students said that the game helped them consolidate the historical facts discussed in class. It is also worth mentioning that participants felt highly engaged in the game without being aware of time passing. Moreover, students' opinions about their in-game collaboration was indicated as positive and almost all students believed that the game helped them collaborate with their partners successfully.

Despite our preliminary positive findings, there is a valid concern that our results associated with game playing may be short-term and due to a 'novelty effect'. Thus, in the future, we plan to conduct a more robustly designed in-depth study to examine HistoryLand impact over an extended period of time. Additionally, we plan to extend the analysis of the results, exploring the effectiveness of the conversational agent scaffolding students' discussions during the collaborative game phase. Lastly, our future work will also focus on enriching the game content to include more historical periods.

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